Program review

App_DoughFactory

5 issue groups · 17 affected source lines

Repeated results are combined below. Every affected line is retained; different values, limits and calculation bases remain separate.

PROGRAM AT A GLANCE	
Saved software	5.24.0
Guidance	CB-Series / e-Series deployment guidance
Installation	default.installation
Source coverage	1,920 script lines; 27 movej, 9 optimovej, 34 movel, 1 optimovel
Needs verification	48 grouped checks need runtime evidence or additional information
Generated	2026-10-30 22:55 UTC

Start here

Review the grouped issues on the next pages. Each entry gives the comparison, affected script lines and any source-estimate assumptions.

Source estimates substitute saved initializers; initialization and runtime values are not proven. UR acceleration recommendations are starting points, not safety limits. Unknown values are not treated as zero or a pass.

Issues to review

01 Acceleration recommendations

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s².

Lines 360, 616, 623, 637, 644, 737, 741, 744, 1747
Source estimate · D9, D10 · UR [1]
Example at line 360: movel(pose_trans(pre_pick_pose, p[0.0, -0.0, PRE_PICK_GAP, 0, 0, 0]), v=vl2*v_scale, a=al2*a_scale)

02 Acceleration recommendations

Configured a=6.25 m/s². Exceeds UR suggested starting value ≤2.5 m/s².

Lines 414
Source estimate · D12 · UR [1]
Example at line 414: movel(pose_add(get_actual_tcp_pose(), p[0, 0, 0.015, 0, 0, 0]), v=vl3*v_scale, a=al3*a_scale)

03 Acceleration recommendations

Configured a=3 m/s². Exceeds UR suggested starting value ≤2.5 m/s².

Lines 448
UR [1]
Example at line 448: movel(retractTo, a=3.0, v=0.1)

04 Acceleration recommendations

Configured a=4.36332 m/s². Exceeds UR suggested starting value ≤2.5 m/s².

Lines 797, 807, 826, 828
Source estimate · D14 · UR [1]
Example at line 797: movel(pose_add(current_pose, p[0, place_y, -0.0, 0, 0, 0]), v=vr2*v_scale, a=ar2*a_scale, r=0.00)

05 Acceleration recommendations

Configured a=2.61799 m/s². Exceeds UR suggested starting value ≤2.5 m/s².

Lines 843, 1835
Source estimate · D7 · UR [1]
Example at line 843: movel(pre_idle_p, v=vr1*v_scale, a=ar1*a_scale, r=0.01)

Independent source review. Not a UR certification or safety validation. Validate runtime behavior and physical setup on the robot.

Checks & source values

Issues already listed above are not repeated here. “Needs verification” means runtime evidence or a specification is missing. “Source value” records saved configuration, not physical correctness.

INITIAL PAYLOAD / COG / INERTIA	
Program context	Source value Saved installation payload: 3.6 kg; CoG=(0.001, 0.0, 0.132) m. URP references ../default. Saved values do not prove the physical tool setup. UR [1]
Lines 3	Source value Mass: 3.6 kg; cog: [0.001, 0.0, 0.132] m from tool mount; inertia: [0.013017, 0.013017, 0.013017, 0.0, 0.0, 0.0] kg·m². UR [5]

DYNAMIC PAYLOAD CHANGES	
Lines 514	Source value Mass: 6.3 kg; cog: [0.0, -0.01, 0.1] m from tool mount; inertia: [0, 0, 0, 0, 0, 0] kg·m². Source estimate · D1 · UR [5]
Lines 799	Source value Mass: 4.7 kg; cog: [0.0, -0.01, 0.1] m from tool mount; inertia: [0, 0, 0, 0, 0, 0] kg·m². Source estimate · D1 · UR [5]
Lines 1535, 1731, 1817	Source value Mass: 4.7 kg; cog: [0.0, -0.01, 0.1] m from tool mount. Source estimate · D1 · UR [1]
Lines 1725, 1811	Source value Mass: 6.3 kg; cog: [0.0, -0.01, 0.1] m from tool mount. Source estimate · D1 · UR [1]

Checks & source values (continued)

DYNAMIC PAYLOAD CHANGES (continued)	
Program context	Needs verification 8 configuration call(s) in source, including definitions and branches. Call counts do not prove updates at every pickup, release or tool change. UR [1]

INITIAL TCP	
Program context	Source value Saved installation TCP dough_factory: (0.0, 0.0, 0.247, -0.0, 3.1414, -0.0). Translation in m, rotation vector in rad. Program set_tcp calls may override this. UR [1]
Lines 24	Source value pose: [0.0, 0.0, 0.247, -0.0, 3.1414, -0.0] (XYZ in m; rotation vector in rad). UR [1]

DYNAMIC TCP CHANGES	
Lines 1536	Source value pose: [0.0, 0.0, 0.247, 0.0, 3.1414181206645937, 0.0] (XYZ in m; rotation vector in rad). Source estimate · D2 · UR [1]
Program context	Needs verification 2 configuration call(s) in source, including definitions and branches. Call counts do not prove updates at every pickup, release or tool change. UR [1]

ACCELERATION RECOMMENDATIONS

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Checks & source values (continued)

ACCELERATION RECOMMENDATIONS (continued)	
Lines 138, 618, 657, 664, 672, 679	Needs verification Configured a=6.10865 rad/s². Within UR suggested starting value ≤13.9626 rad/s². Source estimate · D3 · UR [1]
Lines 156, 158	Needs verification Configured a=0.8 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]
Lines 161, 307	Needs verification Configured a=0.6 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]
Lines 167	Within guideline Configured a=1 m/s². Within UR suggested starting value ≤2.5 m/s². UR [1]
Lines 199	Within guideline Configured a=0.05 m/s². Within UR suggested starting value ≤2.5 m/s². UR [1]
Lines 294, 296, 303, 639	Needs verification Configured a=1 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]
Lines 327, 577, 591, 599, 707, 1758, 1765, 1782, 1787, 1791, 1793	Needs verification Configured a=1.25 m/s². Within UR suggested starting value ≤2.5 m/s². Source estimate · D5 · UR [1]

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Checks & source values (continued)

ACCELERATION RECOMMENDATIONS (continued)	
Lines 329	Needs verification Configured a=2.61799 rad/s². Within UR suggested starting value ≤13.9626 rad/s². Source estimate · D7 · UR [1]
Lines 390	Within guideline Configured a=0.2 m/s². Within UR suggested starting value ≤2.5 m/s². UR [1]
Lines 427	Within guideline Configured a=1.2 m/s². Within UR suggested starting value ≤2.5 m/s². UR [1]
Lines 482	Within guideline Configured a=0.1 m/s². Within UR suggested starting value ≤2.5 m/s². UR [1]
Lines 711, 831	Needs verification Configured a=4.36332 rad/s². Within UR suggested starting value ≤13.9626 rad/s². Source estimate · D14, D15 · UR [1]
Lines 761	Within guideline Configured a=0.3 m/s². Within UR suggested starting value ≤2.5 m/s². UR [1]
Lines 858, 1806	Needs verification Configured a=0.5 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]

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Checks & source values (continued)

ACCELERATION RECOMMENDATIONS (continued)	
Lines 865, 1471, 1473, 1476, 1478, 1481, 1483, 1485, 1487, 1490, 1492, 1494, 1496, 1498, 1500, 1502	Within guideline Configured a=1.39626 rad/s². Within UR suggested starting value ≤13.9626 rad/s². UR [1]
Lines 1801, 1803	Needs verification Configured a=2.44346 rad/s². Within UR suggested starting value ≤13.9626 rad/s². Source estimate · D17 · UR [1]

BLEND RADIUS OVERLAP	
Lines 138, 831, 1782, 1787, 1791	Needs verification Configured radius r=0.02 m. UR [4]
Lines 156, 618, 1801, 1803	Needs verification Configured radius r=0.1 m. UR [4]
Lines 158, 294, 639	Needs verification Configured radius r=0.15 m. UR [4]
Lines 161, 167, 199, 307, 327, 360, 390, 414, 427, 448, 482, 577, 591, 707, 711, 744, 761, 797, 865, 1471, 1473, 1476, 1478, 1481, 1483, 1485, 1487, 1490, 1494, 1496, 1498, 1500, 1502, 1747, 1806, 1835	Source value Configured radius r=0 m. UR [4]

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Checks & source values (continued)

BLEND RADIUS OVERLAP (continued)	
Lines 296	Needs verification Configured radius r=0.25 m. UR [4]
Lines 303	Needs verification Configured radius r=0.12 m. UR [4]
Lines 329, 843, 1793	Needs verification Configured radius r=0.01 m. UR [4]
Lines 599	Needs verification Configured radius r=0.003 m. UR [4]
Lines 616, 637	Needs verification Configured radius r=0.08 m. UR [4]
Lines 623, 644, 657, 664, 672, 679, 737, 741, 858, 1492, 1758, 1765	Needs verification Configured radius r=0.05 m. UR [4]
Lines 807	Needs verification Configured radius r=0.0025 m. UR [4]
Lines 826	Needs verification Configured radius r=0.0025 m. Outgoing move distance=0.0469574 m. UR [4]
Lines 828	Needs verification Configured radius r=0.0025 m. Incoming move distance=0.0469574 m. Adjacent radii total=0.005 m. UR [4]

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Checks & source values (continued)

BLEND RADIUS OVERLAP (continued)	
Program context	Needs verification line 22: motion_version_set(2) Version 2 can reduce blend radii during planning. A static geometric overlap must not be presented as a proven skipped waypoint. UR [3]

BLENDED MOTION CONTINUATION	
Program context	Needs verification 35 motion calls have nonzero or unresolved radii. Verify continuation through branches, waits, stops and helper functions on the applicable controller. Static source inspection does not establish the executed path. UR [4]

VELOCITY SPECIFICATIONS	
Program context	Needs verification Joint and tool speed limits depend on the configured Normal/Reduced safety modes and application. Saved installation safety serialization is not interpreted as an active speed specification. Advanced overrides can supply verified limits. UR [10]
Lines 138, 618, 657, 664, 672, 679, 1801, 1803	Needs verification Configured v=9.77384 rad/s. Enter the applicable joint velocity specification to check this value. Source estimate · D4 · UR [1]
Lines 156, 158	Needs verification Configured v=0.85 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]

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Checks & source values (continued)

VELOCITY SPECIFICATIONS (continued)	
Lines 161	Needs verification Configured v=0.7 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]
Lines 167	Needs verification Configured v=1 m/s. Enter the applicable TCP velocity specification to check this value. UR [1]
Lines 199	Needs verification Configured v=0.05 m/s. Enter the applicable TCP velocity specification to check this value. UR [1]
Lines 294, 296, 303, 639	Needs verification Configured v=1 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]
Lines 307, 1806	Needs verification Configured v=0.5 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]
Lines 327, 577, 591, 599, 707, 1758, 1765, 1782, 1787, 1791, 1793	Needs verification Configured v=2 m/s. Enter the applicable TCP velocity specification to check this value. Source estimate · D6 · UR [1]
Lines 329	Needs verification Configured v=4.18879 rad/s. Enter the applicable joint velocity specification to check this value. Source estimate · D8 · UR [1]

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Checks & source values (continued)

VELOCITY SPECIFICATIONS (continued)	
Lines 360, 616, 623, 637, 644, 737, 741, 744, 1747	Needs verification Configured v=5 m/s. Enter the applicable TCP velocity specification to check this value. Source estimate · D11 · UR [1]
Lines 390	Needs verification Configured v=0.2 m/s. Enter the applicable TCP velocity specification to check this value. UR [1]
Lines 414	Needs verification Configured v=10 m/s. Enter the applicable TCP velocity specification to check this value. Source estimate · D13 · UR [1]
Lines 427	Needs verification Configured v=0.25 m/s. Enter the applicable TCP velocity specification to check this value. UR [1]
Lines 448	Needs verification Configured v=0.1 m/s. Enter the applicable TCP velocity specification to check this value. UR [1]
Lines 482	Needs verification Configured v=1.5 m/s. Enter the applicable TCP velocity specification to check this value. UR [1]
Lines 711, 831	Needs verification Configured v=6.98132 rad/s. Enter the applicable joint velocity specification to check this value. Source estimate · D16 · UR [1]
Lines 761	Needs verification Configured v=0.3 m/s. Enter the applicable TCP velocity specification to check this value. UR [1]

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Checks & source values (continued)

VELOCITY SPECIFICATIONS (continued)	
Lines 797, 807, 826, 828	Needs verification Configured v=6.98132 m/s. Enter the applicable TCP velocity specification to check this value. Source estimate · D16 · UR [1]
Lines 843, 1835	Needs verification Configured v=4.18879 m/s. Enter the applicable TCP velocity specification to check this value. Source estimate · D8 · UR [1]
Lines 858	Needs verification Configured v=0.4 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established. UR [6]
Lines 865, 1471, 1473, 1476, 1478, 1481, 1483, 1485, 1487, 1490, 1492, 1494, 1496, 1498, 1500, 1502	Needs verification Configured v=1.0472 rad/s. Enter the applicable joint velocity specification to check this value. UR [1]

MOVE DISTANCES

Checks & source values (continued)

MOVE DISTANCES (continued)	
Lines 138, 156, 158, 161, 167, 199, 294, 296, 303, 307, 327, 329, 360, 390, 414, 427, 448, 482, 577, 591, 599, 616, 618, 623, 637, 639, 644, 657, 664, 672, 679, 707, 711, 737, 741, 744, 761, 797, 807, 826, 831, 843, 858, 865, 1471, 1473, 1476, 1478, 1481, 1483, 1485, 1487, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1747, 1758, 1765, 1782, 1787, 1791, 1793, 1801, 1803, 1806, 1835	Needs verification Move distance unresolved: initial pose, joint-space trajectory, target/frame expression or intervening control flow is unknown. UR [4]
Lines 828	Source value Unblended move distance=0.0469574 m from preceding source waypoint (translation only). Derived from a rigid offset of the same named pose; runtime pose stability and execution require verification. Source-definition substitution is conditional on initialization and unchanged values. UR [4]
Program context	Source value Known source path distance=0.0469574 m across 1 of 71 move(s). This sums resolvable unblended translational segments only; unknown segments, first approach, rotations, branches and repetitions are excluded. It is not total executed travel. UR [4]

Source notes

How to interpret the review

This is a static source review; no URScript is executed. Counts include definitions and branches, not execution frequency. Source values do not establish measured speeds or accelerations. Blends depend on controller planning; a geometric warning does not prove a skipped waypoint. Validate physical payload, centre of gravity, inertia and TCP, including each workpiece or tool change.

D references identify saved initializer substitutions. They are conditional estimates, not proof that those values were initialized or unchanged when a move ran.

INITIALIZER EVIDENCE	
D1	PAYLOADS (line 1261) Used at script lines 514, 799, 1535, 1725, 1731, 1811, 1817
D2	TCPS (line 1272) Used at script lines 1536
D3	a_scale=2.5 (line 1278), ar3=2.44346095 (line 1297) Used at script lines 138, 618, 657, 664, 672, 679
D4	v_scale=4 (line 1280), vr3=2.44346095 (line 1289) Used at script lines 138, 618, 657, 664, 672, 679, 1801, 1803
D5	a_scale=2.5 (line 1278), al1=0.5 (line 1291) Used at script lines 327, 577, 591, 599, 707, 1758, 1765, 1782, 1787, 1791, 1793
D6	v_scale=4 (line 1280), vl1=0.5 (line 1283) Used at script lines 327, 577, 591, 599, 707, 1758, 1765, 1782, 1787, 1791, 1793
D7	a_scale=2.5 (line 1278), ar1=1.04719755 (line 1295) Used at script lines 329, 843, 1835
D8	v_scale=4 (line 1280), vr1=1.04719755 (line 1287) Used at script lines 329, 843, 1835
D9	a_scale=2.5 (line 1278), al2=1.25 (line 1292) Used at script lines 360, 616, 623, 637, 644, 744, 1747

Source notes (continued)

INITIALIZER EVIDENCE (continued)	
D10	a_scale=2.5 (line 1278), vl2=1.25 (line 1284) Used at script lines 737, 741
D11	v_scale=4 (line 1280), vl2=1.25 (line 1284) Used at script lines 360, 616, 623, 637, 644, 737, 741, 744, 1747
D12	a_scale=2.5 (line 1278), al3=2.5 (line 1293) Used at script lines 414
D13	v_scale=4 (line 1280), vl3=2.5 (line 1285) Used at script lines 414
D14	a_scale=2.5 (line 1278), ar2=1.74532925 (line 1296) Used at script lines 711, 797, 807, 826, 828
D15	a_scale=2.5 (line 1278), vr2=1.74532925 (line 1288) Used at script lines 831
D16	v_scale=4 (line 1280), vr2=1.74532925 (line 1288) Used at script lines 711, 797, 807, 826, 828, 831
D17	ar3=2.44346095 (line 1297) Used at script lines 1801, 1803

Reviewed source files

URP references ../default

/home/apex/workspace/motionSim/tests/full-backup.tar :: /programs/App_DoughFactory/App_DoughFactory.urp
SHA-256: 81e303f2b98254bd259a53cf03e0961e112d9ee66d7a67f3e2194b925c50133b

/home/apex/workspace/motionSim/tests/full-backup.tar :: /programs/App_DoughFactory/App_DoughFactory.script
SHA-256: 7366e9a00551a3459df57b787d1cfdb21ce15b62fc0d02926533fbd25f89cd5e

/home/apex/workspace/motionSim/tests/full-backup.tar :: /programs/default.installation
SHA-256: 37475581ef1f1a022997dbc9ba9622de43e4da4355a91d7c00b5dfd90879869e

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Source notes (continued)

UR references

[1] UR Important Deployment Points (CB-Series/e-Series)
https://www.universal-robots.com/articles/ur/robot-care-maintenance/important-deployment-points/

[3] UR Motion Version (PolyScope 5.22+)
https://www.universal-robots.com/manuals/EN/HTML/SW5_22/Content/prod-scriptmanual/all_scripts/motion_version_desc.htm

[4] UR Blend Parameters
https://www.universal-robots.com/manuals/EN/HTML/SW5_19/Content/prod-usr-man/software/PolyScope/content/BasicProgNodes/commandtab_way_blend_para.htm

[5] UR Payload configuration (PolyScope 5.24)
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/set_target_payload.htm

[6] UR OptiMove parameter ranges
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/movej_optimove.htm

[10] UR configurable joint and tool speed limits
https://www.universal-robots.com/manuals/EN/HTML/SW5_25/Content/prod-usr-man/complianceUR7e/H_g5_sections/safetyFunctionsAndinterfaces/configurable_functions.htm

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